

Spectral Radius Minimization for Signed Squares of Cycles

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Abstract

Keywords: [KEYWORDS]

MSC 2020: [PRIMARY MSC]; [SECONDARY MSC]

1 Introduction

2 Switching Coordinates and the Reference Phase

2.1 Edge signings, switching, and Hamilton gauge

2.2 Flux words, holonomy, and the attaining candidate

2.3 The squared local operator

2.4 The period-eight reference phase

3 Gaps, Charges, and Translation Sectors

3.1 Cyclic gaps and excess charge

3.2 Translation sectors

3.3 Composition and legal cyclic closure

4 The Elementary Six-Gap Phase Slip

4.1 The bilateral interface and its essential spectrum

4.2 Algebraic candidate and physical matching

4.3 The global G6 edge

4.4 Rank-two symmetry

5 Optimality Among Single-Gap Interfaces

5.1 Canonical single-gap operators

5.2 Exact finite-gap comparisons

5.3 Uniform control of the large gaps

5.4 Reference and single-gap hierarchy

6 Phase Slips on Finite Rings

6.1 Legal residue constructions

6.2 Identification of local patches

6.3 Discrete IMS localization

6.4 Residue upper bounds and the analytic tail

6.5 Structural refinement for separated interfaces

7 Finite Completion of the Classification

7.1 Exact-computation protocol

7.2 Orders 8 through 32

7.3 Orders 34 through 46

7.4 The exact bridge from 48 to 238

7.5 Exhaustive synthesis

8 Concluding Remarks

Data and Code Availability

A Exact Spectral Certification of the Reference Phase and G6

B Exact Finite Classification and Completeness

References

- [1] Yonatan Bilu and Nathan Linial. Lifts, discrepancy and nearly optimal spectral gap. *Combinatorica*, 26(5):495–519, 2006.